

SEBASTIEN Coïc

M2 Quant Finance Engineer | Seeking V.I.E/CDI in Front Office (Trading, Structu, Quant) or Risk – Available from Nov. 2026 onwards

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Professional Experiences

Natixis Corporate & Investment Banking

Apr 2025 – Oct 2025

Quantitative Trading Analyst – XVA Modelling

Paris, France

- Pricing of Swaps, XCCY Swaps & Swaptions: Fast and accurate valuation using Black-Scholes, Heston, and SABR; implementation of multi-curve and multi-currency pricing frameworks. Developed CVA engines under LGM-1F, leveraging GPR and Bayesian Quadrature for sensitivity analysis (Delta, Gamma, Vega, Theta, Cega).
- Stress Testing & Scenario Modeling: Development of systemic shock simulations using copulas, EVT, and Monte Carlo methods; application of unsupervised learning (k-means, DBSCAN, GMM) to segment exposures and enhance stress testing accuracy across trading positions. Risk & XVA Calibration: Calibration of market risk models; impact analysis on CET1, VaR, and ES for interest rate derivatives.

Spread Research

Apr 2024 – Oct 2024

Convertible Bonds, Fixed Income Quantitative Analyst & Researcher

Paris, France

- Bond Pricing & Risk Tools: Built and automated a pricing engine for convertible and high-yield bonds (Greeks, Z-Spread, OAS, implied vol) using VBA and Bloomberg/FactSet, enabling sharp valuation and real-time risk tracking.
- Market Views & Trade Ideas: Delivered daily/weekly reports with actionable insights (ICS, parity, Monte Carlo spreads); formulated underweight/overweight strategies based on volatility signals, hedging ratios, and scenario-driven backtests.

Société Générale Corporate & Investment Banking

Apr 2023 – Oct 2023

Portfolio Valuation Officer

Paris La Défense, France

- Fund Valuation & NAV Controls: Advanced pricing models engineered for mutual funds, incorporating Monte Carlo simulations and dynamic volatility adjustments; enhanced NAV monitoring via scenario-based stress testing and automated PnL consistency checks.
- Process Automation & Risk Reporting: Optimized order flow and audit trails using VBA; standardized communication with asset managers and custodians to ensure timely, regulator-ready reporting (AMF-compliant).

Education

Financial Mathematics - Dual Degree: Engineering/Master | CY TECH (ex EISTI)

2022 – 2025

(Ranked 1/103)

Paris, France

- Focus: Monte Carlo Simulations (Black-Scholes model); Stochastic Processes; Machine Learning & Data Science; Portfolio Management & Financial Risk (VaR, CVaR and TVaR); Calibration; Advanced Numerical Methods; Statistics; Econometrics; Contingent Claim Valuation; Advanced Probability; Actuarial Science; Microeconomics and Macroeconomics; Topology.

Maths Sup/Maths Spé Preparatory Class | CY TECH (ex EISTI)

2020 – 2022

Option Mathematics/Physics (Ranked 2/216)

Paris, France

Quantitative Projects & Research (Available on my LinkedIn/WebSite)

Course of Advanced Portfolio Theory: From Value at Risk to Merton's Framework

2025

- Portfolio optimization via convex duality in incomplete markets using $\hat{X} = I(\lambda\xi/S_T^0)$.
- Theoretical critique of VaR; implementation of AVaR as a coherent, subadditive risk measure.
- Characterization of optimal strategies through equivalent martingale measures $\mathcal{M}_e$.

Calibration of the Heston Model

2024

- Implementation and calibration of the Heston model for option pricing.
- Application of the Levenberg-Marquardt method for fast and accurate calibration.
- Estimation of market parameters and fitting of the implied volatility surface.

Systemic Risks and Contagion in the Financial Network

2022

- Stochastic modeling of contagion and shocks: Simulation of interbank default cascades and non-linear loss propagation using stochastic differential equations with leverage effects.
- Monte Carlo and network analysis: Estimation of default probabilities, VaR/CVaR, and identification of critical nodes in interbank networks under extreme scenarios.

Certifications

CFA Candidate

2025

Bloomberg: Market Concept Certificate, Finance Fundamentals Certificate, Environmental Governance Certificate.

2024

JP Morgan: Investment Banking Virtual Experience Program Certificate.

2023

Goldman Sachs: Excel Skills for Business Experience Program.

2023

AMF Certificate

2023

Others

French: Native **English:** Fluent **Spanish:** Elementary

Computer skills: Bloomberg, FactSet, LaTeX, Pack Office, RStudio, Git, Github, Maple.

Programming Languages: Python (*Advanced*), VBA Excel (*Advanced*), R (*Intermediate*), C/C++ (*Intermediate*).

Hobbies: Tennis ex-french national player (*Ranked 5/6*), Surf, Mathematics Instructor (*CPGE*) & Oral Examiner.